Fiscal Year:	FY 2004	**Task Last Updated:**	FY 06/22/2005
PI Name:	Maurer, Richard H. Ph.D.		
Project Title:	Neutron Energy Spectrometer Flight Experiments		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Technology Development Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Laurel	**State:**	MD
Zip Code:	20723-6099	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
Start Date:	10/01/2000	**End Date:**	03/31/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Fainchtein, Raul (The Johns Hopkins University) Dicello, John (Johns Hopkins Cancer Center) Gold, Robert (The Johns Hopkins University Applied Physics Laboratory) Goldsten, John (The Johns Hopkins University Applied Physics Laboratory) Kinnison, James (The Johns Hopkins University Applied Physics Laboratory) Roth, David (The Johns Hopkins University Applied Physics Laboratory)		
Grant/Contract No.:	NCC 9-58-TD00204		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	High-energy charged particles of extra-galactic, galactic and solar origin collide with spacecraft structures in Earth orbit outside the atmosphere and in interplanetary travel beyond the Earth's magnetosphere. These primaries create a number of secondary particles inside the structures that can produce a significant ionizing radiation environment. This radiation is a threat to long term inhabitants for space missions and produces an increased risk of cancer, CNS and DNA damage. The primary high energy cosmic rays and trapped protons collide with spacecraft materials such as aluminum and silicon and create secondary particles inside structures that are charged particle fragments and neutrons. The effect of tens of grams per square centimeter of structure or atmosphere is to convert and multiply the primary proton "beam" into a secondary environment dominated by neutrons. Charged protons are readily detected and instruments are already in existence for this task. Neutrons are electrically neutral and much more difficult detect. These neutrons are estimated to contribute 10-30% of the dose inside space structures and cannot be ignored. Currently there is no compact, portable and real time neutron detector instrumentation available for use inside spacecraft or on planetary surfaces. As a product of our previous NSBRI funding during FY 1998-2001 we had met the original aims of the project to design and fabricate an engineering prototype neutron spectrometer that was flown on F-15 and F-18 aircraft flights from NASA Dryden Flight Center. The spectrometer consists of both low and high energy subsystems. The detection of low energy neutrons (0.025 eV-1 MeV) is accomplished using a conventional helium 3 gas tube. The detection of high energy neutrons (5-800 MeV) is achieved using a 5 mm thick lithium drifted silicon solid state device. The neutron spectrometer was flown on two flights 13-14 August 2001 in a pod under the wing of an F-18. A third successful flight in the same pod under the fuselage of an F-15 was executed in October 2001. The main positive result from the three flights was the verification of our engineering design and not the limited data obtained due to the short duration (~2 hours) of the aircraft flights. The value for our hardware was the proven approach in handling high voltage at high altitude corona region that will be employed for balloon flights. Efforts in FY 2002 and 2003 were directed at designing and fabricating a neutron spectrometer for high altitude balloon flights. The electronics were made more robust and compact for the balloon flight instrument. The detector suite was changed to include a Medium Energy Spectrometer (MES) for the fast neutrons in the 1-20 MeV energy range in addition to the thick silicon detector for the > 20 MeV neutron energies. The helium 3 tube was not included for the low energy neutrons (10 keV-1 MeV) since this system was validated on the aircraft flights and will be readily available for flight efforts. The MES is a Bicron 454 plastic scintillator detector system that borrows from the development of a similar system for the APL unmanned MESSENGER mission to Mercury. Sophisticated energy deposition signal time discrimination allows the observation of both scattering and capture peaks of the neutrons in the Bicron detector for individual counts and energies. Development of experience in the calibration and use of this detector system was one of the interim goals of the project. Balloon flights were executed from Fort Sumner, NM at an altitude of 85,000 feet on two occasions--October 9, 2002 and October 9, 2003. The altitude of 85,000 feet was chosen since the amount of atmosphere remaining (~20 grams per square centimeter) is the same as the amount of carbon dioxide at the surface of Mars and should yield a reasonable simulation of the downward neutron spectrum on Mars. The October 2002 flight did not yield any useful scientific data due to engineering problems with the high voltage connection to the silcon detector and ground loop issues between the electronics and the aluminum container of the instrument. The problems were corrected during FY 03 and the October 2003 flight yielded useful scientific data. Initial analysis showed that we obtained a highly moderated neutron energy spectrum with the majority of of neutrons in the energy range between 20 and 35 MeV. Modeling of the detector shielding geometry is necessary to deduce that on the lid of the instrument relative to that measured at the detector location. This task is our starting point in FY 05 under our new NSBRI grant. In January 2000 we were notified that our proposal titled "Development of a Neutron Spectrometer to Assess Biological Radiation Damage Behind Spacecraft Materials" submitted in March 1999 in response to NASA NRA 98-Heds-05 would be funded for a period of 3.5 years from May 2000 to November 2003. For the evaluation of spacecraft structural and shielding materials we built a stack detector version of the neutron spectrometer compatible with ground-based accelerator research. We verified its successful operation at Columbia University's RARAF in November 2001, and then proceeded with spacecraft shielding experiments using 200 MeV proton beams at the Indiana University Cyclotron Facility in November 2002 and November 2003 and 500 MeV proton beam at TRIUMF in Vancouver, Canada in September 2003. Aluminum, carbon and polyethylene block targets were used to simulate spacecraft materials. The results yielded neutron production energy spectra showing the reduced yield from carbon based materials and the moderating (scattering) effects of polyethylene when compared to aluminum. Our data validate the recent conclusion about aluminum being the least suitable spacecraft material with respect to increased neutorn production and human radiation effects. This was the last year for this project.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The main earth-based application for this project/instrument development is related to homeland security and the monitoring of freight/cargo for the presence of fast and high energy neutrons from clandestine radioactive material.
	We published our basic spectrum inversion methodology in the February 2003 article in Radiation Research titled "High Energy Neutron Spectroscopy with Thick Silicon Detectors". The methodology is based on measurement of the detector response function for high energy neutrons and inversion of this response function with measured deposition data to deduce neutron energy spectra. Also in February 2003 our concept for a Mars Lander neutron spectrometer was published in an Acta Astronautica article titled "MArs Neutron Energy Spectrometer (MANES): an instrument for the Mars 2003 Lander". Although this mission was cancelled, our efforts in conceptual design resulted in this publication. During the January-April 2003 time period we improved our balloon flight instrument design and made it more robust and reliable. We qualified it for flight at the NSBF facility at Fort Sumner, NM in May 2003 and then had to wait until October 9, 2003 to execute our flight due to internal NASA bureaucratic snags with approval of our flight application and agreement. The October 2003 flight at 85,000 feet (an altitude above which there remains about 20 grams per square centimeter of atmosphere that is about the same as the atmospheric thickness on the surface of Mars) for approximately 20 hours of float time yielded useful scientific data from the thick silicon detector for high energy neutrons. Initial

Task Progress:	analysis indicates a highly moderated integral neutron spectrum due to the inherent shielding of our aluminum instrument package and the one atmosphere of air pressure inside the container as well as the Cesium Iodide charged particle discrimination scintillator that surrounds the silicon detector. we are presently in the process of modeling the shielding configuration in order to deduce the spectrum at the top lid of the container from the moderated one at the detector. Spacecraft shielding materials experiments were carried out at the TRIUMF accelerator facility in Vancouver, Canada with 500 MeV protons in September 2003 and at the Indiana University Cyclotron with 200 MeV protons in November 2003. The proton beams collided with thick targets of aluminum, carbon and polyethylene, and the neutrons produced at angles between 0 and 90 degrees with respect to the beam direction were monitored. The results of these accelerator experiments were presented at the International Conference on Radiation Shielding on Madeira Island in May 2004. For the three years of our funded project we basically advanced the TRL of the neutron energy spectrometer instrument to a level of 6 by designing, fabricating, qualifying and executing both high altitude aircraft and balloon flights. Enough data was acquired on the second balloon for a statistically accurate measurement of the downward high energy neutron spectrum at 85,000 feet altitude.
Bibliography Type:	Description: (Last Updated: 04/27/2011)
Articles in Peer-reviewed Journals	Kinnison JD, Maurer RH, Roth DR, Haight RC. "High-energy neutron spectroscopy with thick silicon detectors." Radiat Res. 2003 Feb;159(2):154-60. PMID: 12537520 , Feb-2003
Articles in Peer-reviewed Journals	Kinnison JD, Maurer RH, Roth DR, McNulty PJ, Abdel-Kader WG. "Neutron-induced pion production in silicon-based circuits." IEEE Transactions on Nuclear Science. 2003 Dec;50(6):2251-5. http://dx.doi.org/10.1109/TNS.2003.821385 , Dec-2003
Articles in Peer-reviewed Journals	Maurer RH, Kinnison JD, Roth DR. "Neutron production from 200-500 MeV proton interaction with spacecraft materials." Radiat Prot Dosimetry. 2005;116(1-4 Pt 2):125-30. PMID: 16604612 , Feb-2005
Articles in Peer-reviewed Journals	Maurer RH, Roth DR, Kinnison JD, Goldsten JO, Gold RE, Fainchtein R. "MArs Neutron Energy Spectrometer (MANES): an instrument for the Mars 2003 Lander." Acta Astronaut. 2003 Jan-Mar;52(2-6):405-10. PMID: 12674102 , Mar-2003
Presentation	Maurer, R. H., Kinnison, J. D. and Roth, D. R. "Neutron Energy Spectra from 200 MeV Proton Interaction with Spacecraft Materials " Jul-2003
Presentation	Maurer, R. H., Kinnison, J. D. and Roth, D. R. "Neutron Production from 200-500 MeV Proton Interaction with Spacecraft Materials " May-2004